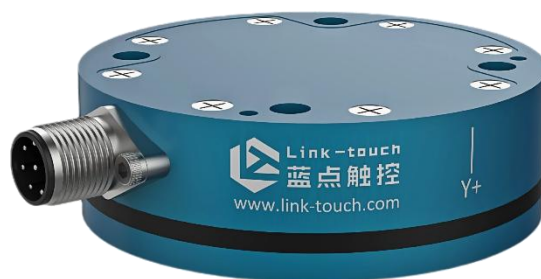




6-dimensional Force and Moment Sensor



Focus on high precision force measurement and control
The pioneer in industry of multi-dimensional Force and Moment sensor

Engineered Products for Sensor Productivity

Version: 1.5.6

Foreword

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This manual is periodically updated to align with the latest functions of products. Our company will not be responsible for any errors or incomplete information in this document. Our company reserves all copy and patent rights.

Considering that our products are applied to robot or automatic control field, we suggest stopping use of products until the errors and malfunctions are fixed to prevent any injury to life of people.

Anyone who uses our products in the systems that are potentially harmful to human life shall get our written approval and is authorized to use our products to prevent cause to injury or death of people.

Note:

Please read the manual thoroughly prior to calling our service office. Pay attention to the following information before calling.

1. Sensor model number (e.g.ST-6-200-10-RS-1805012)
2. Correct and detailed description of problems
3. Hardware and software information: Operating system, drivers, application software, and other relevant information about your configuration.

If possible, be closed to the sensor while calling

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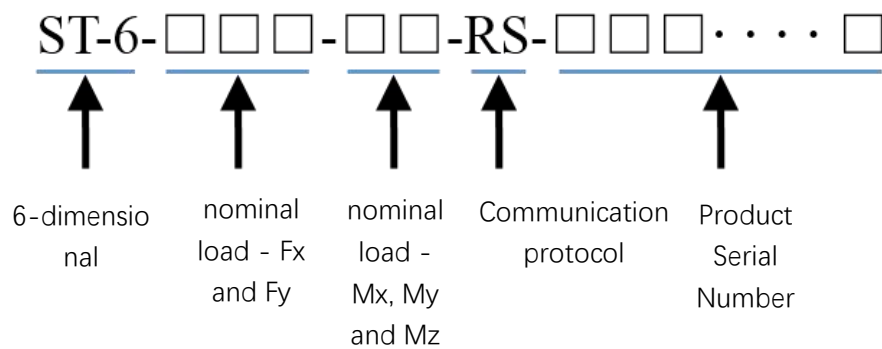
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Glossary

Term	Definition
Accuracy	Reflects the proximity of the measurement result to the true value, called the accuracy, which corresponds to the magnitude of the error. Therefore, the magnitude of error can be used to indicate the accuracy. The smaller, the accuracy higher.
Resolution	The System can measure the minimum amount of change in load, which is usually much less than accuracy.
Calibration	To determine the function of the sensor under specified conditions, use a known mass of the object to be calibrated.
Recalibration	Regularly calibrate measuring equipment such as sensor and voltmeter to prove that it is still measured correctly. If the measurement is not correct, the device should be calibrated.
Product Certificate	Declare that the equipment has been tested in accordance with national standards and meets the national standards in the application field.
Compound Loading	Any load that is not purely in one axis.
Coordinate Frame	Set coordinate origin and motion calculation reference.
DoF	Degrees of freedom.
Rs-422 Bus	RS-422 serial communication bus supports the transmission of one-to-many differential data, with strong anti-noise ability and long transmission distance.
RS422 Communication Protocol	In RS-422 the convention for data transfer control.
Force and Moment	Force/Moment
Force and Moment Transducer	The device that converts force and moment into an electrical signal
Range	Supports the extreme values of force and moment measurements when the sensor is in normal use.
Full-Scale Error	A measurement of sensing error. For example, if the calibrated measurement range of a sensor is 100N and the sensor is accurate to within 1N, that sensor will have a Full-Scale Error of 1%(1%=0.01=1N/100N)。
Hysteresis	A source of measurement error caused by the residual effects of previously applied loads.
IP65	Protection level of foreign equipment intrusion of electrical equipment: 6 - completely prevent dust from entering 5 - low pressure injection at any angle has no effect.
MAX.	The largest amount of load in a single axis (all other axes are unloaded) that the

Single-Axis Overload	transducer can withstand without damage
Error	The maximum expected error that can occur when the sensor is in normal use.
Offset Compensation	Correction of errors that change the zero point of a transducer's readings.
Overload	The condition where more load is applied to the transducer. This will result in saturation.
ADC	Analog-to-digital converter.
6-D input	Fx, Fy, Fz, Mx, My, and Mz.
Tool Side	The side used to mount user-defined tool or tool adapter.
Fixed Side	The side used to fix the sensor.
M_{xy}	The resultant moment vector comprised of components Tx and Ty
UDINT	32 bit unsigned integer
UINT	16 bit unsigned integer
USINT	8 bit unsigned integer
DINT	32 bit signed integer



1. Safety

The safety section describes general safety guidelines for the product(s), an explanation of the notifications is found in this manual, and the safety precautions applicable to the product(s). More specific notifications are embedded within the sections of the manual where they apply.

1.1 Explanation of Notifications

The notifications included here are specific to the product(s) covered by this manual. The user should heed all notifications from the robot manufacturer and/or the manufacturers of other components used in the installation



DANGER: Notification of information or instructions that if not followed will result in death or serious injury. The notification provides information about the nature of the hazardous situation, the consequences of not avoiding the hazard, and the method for avoiding the situation



WARNING: Notification of information or instructions that if not followed could result in death or serious injury. The notification provides information about the nature of the hazardous situation, the consequences of not avoiding the hazard, and the method for avoiding the situation.



CAUTION: Notification of information or instructions that if not followed could result in moderate injury or cause damage to equipment. The notification provides information about the nature of the hazardous situation, the consequences of not avoiding the hazard, and the method for avoiding the situation

NOTICE: Notification of specific information or instructions about maintaining, operating, installation, or setup of the product that if not followed could result in damage to equipment. The notification can emphasize but is not limited to specific grease types, good operating practices, or maintenance tips.

1.2 General Safety Guidelines

The customer should confirm that the nominal loads are less than maximum loads as expected during operation. Since static load is less than dynamic loads, it's necessary to pay attention to the dynamic loads.

1.3 Safety Precautions



WARNING: Performing maintenance or repair on the sensor, while circuits (e.g. power, water, and air) are energized could result in serious injury. Discharge and verify all energized circuits are de-energized in accordance with the customer's safety practices and policies.



CAUTION: Modifying or disassembly of the sensor could cause damage and void the warranty. Users should use the installation components provided by our company to install and debug according to the operating manual specifications.



CAUTION: Using fasteners that exceed the customer interface depth penetrates the body of the sensor, damages the electronics, and voids the warranty. Installation method and component selection need to be reasonable.



CAUTION: Do not overload the transducer. Exceeding the single-axis overload values of the transducer, causes irreparable damage.



CAUTION: Do not contact the IP65 seal. Contacting the seal can cause the sensor to malfunction.



CAUTION: The sensor should be protected from impact and shock loads that exceed rated ranges during transport as the impacts can damage the sensor's



CAUTION: Avoid damage to the sensor from Electro-Static Discharge. Ensure proper grounding procedures are followed when handling the sensor or cables connected to the sensor. Failure to follow proper grounding procedures could damage the sensor.



CAUTION: Damage to the sensor can cause damage to equipment inside the sensor unit. The user should avoid damage to the sensor's casing.

2. Product Overview

WRIST Series Sensor System can measure three components of force (F_x , F_y , F_z) and moment (M_x , M_y , M_z) and send the measurements to user's equipment in real time via cables. The sensor connects the user's equipment on one side and the user's tool fixture on another side.

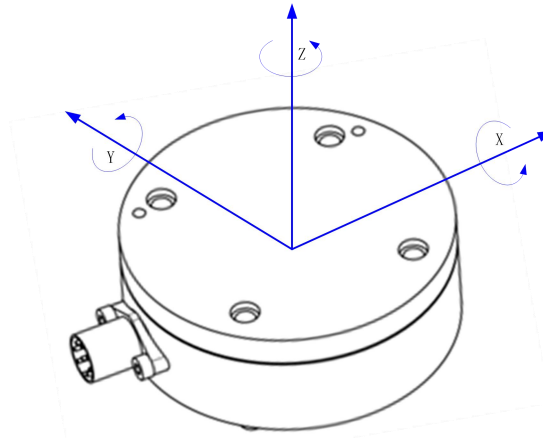


Fig. 2.1: Coordinate System for the Sensor

The fig 2.1 illustrates coordinate system for the sensor with the following features:

1. Output of 6 components of force and moment with high accuracy.
2. Tailored for the robots on production lines
3. Built-in temperature compensation unit
4. High resolution of measurement
5. High bandwidth of response

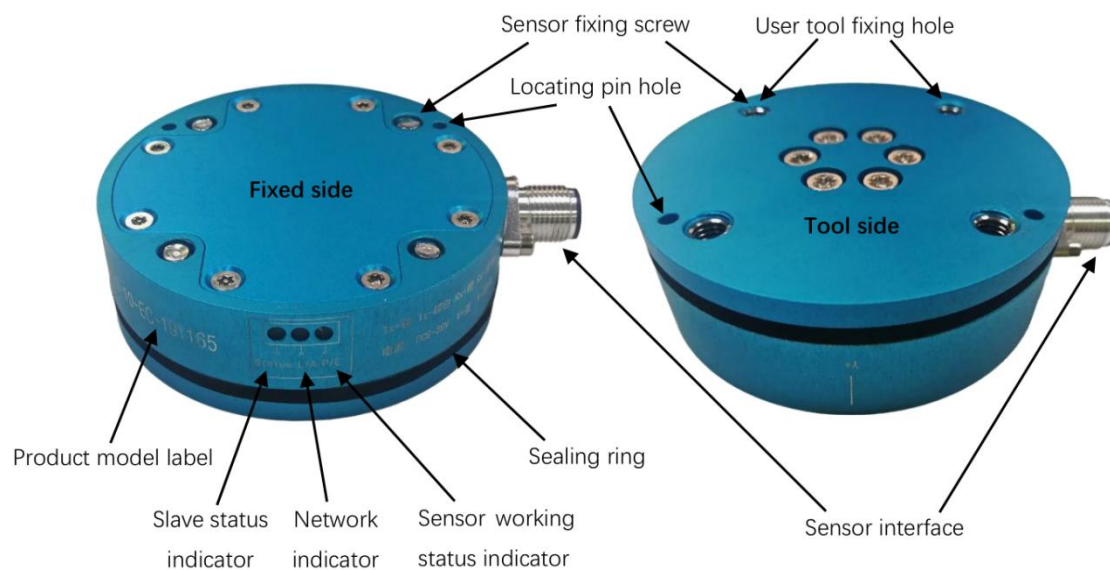


Fig. 2.2: The WRIST Sensor

3. Installation



WARNING: Performing maintenance or repair on the sensor when circuits (e.g. power, water, and air) are energized could result in death or serious injury. Discharge and verify all energized circuits are deenergized in accordance with the customer's safety practices and policies.



CAUTION: Modification or disassembly of the sensor could cause damage and void the warranty. Users should use the installation components provided by our company to install and debug based on the operating manual specifications.



CAUTION: Using fasteners that exceed the customer interface depth penetrates the body of the sensor, damages the electronics, and voids the warranty.



CAUTION: Damage to the sensor shell can cause damage to equipment inside the sensor unit. The user should avoid damage to the sensor shell.



CAUTION: The sensor's single-axis overload will cause irreparable damage to the sensor itself.



CAUTION: Do not touch the ip65 seals at will, which may cause the sensor to malfunction.



CAUTION: Avoid damage to the sensor from Electro-Static Discharge. Ensure proper grounding procedures are followed when handling the sensor or cables connected to the sensor. Failure to follow proper grounding procedures could damage the sensor.



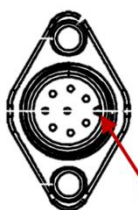
CAUTION: The sensor should avoid shock and shock loads beyond its tolerance during transport, as the impact can damage the internal structure of the sensor and cause its performance to deteriorate.



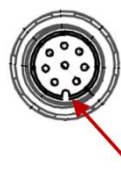
CAUTION: Thread locker applied to fasteners must not be used more than once. Fasteners may become loose and cause equipment damage. Always apply new thread locker when reusing fasteners.



CAUTION: Do not apply excessive force to the sensor and cable connector during installation, or damage will occur to the connectors. Align the keyway on the sensor and cable connector during installation to avoid applying excessive force to the connectors.



Key



Slot

3.1 Installation Instruction

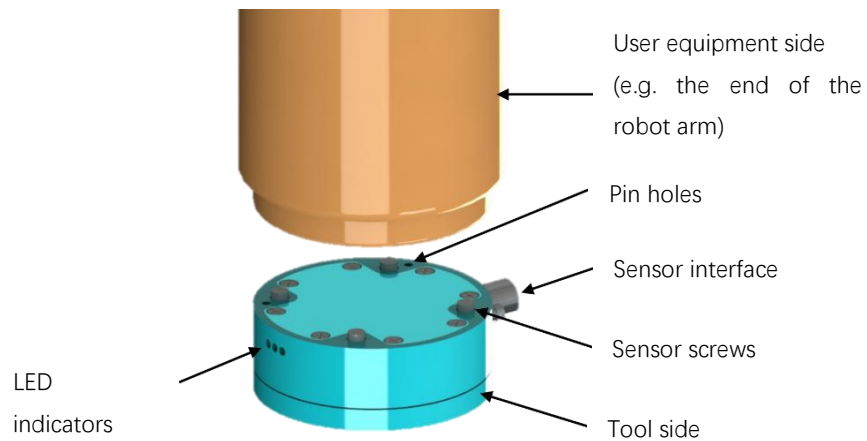


Fig. 3.1: Sensor Installation

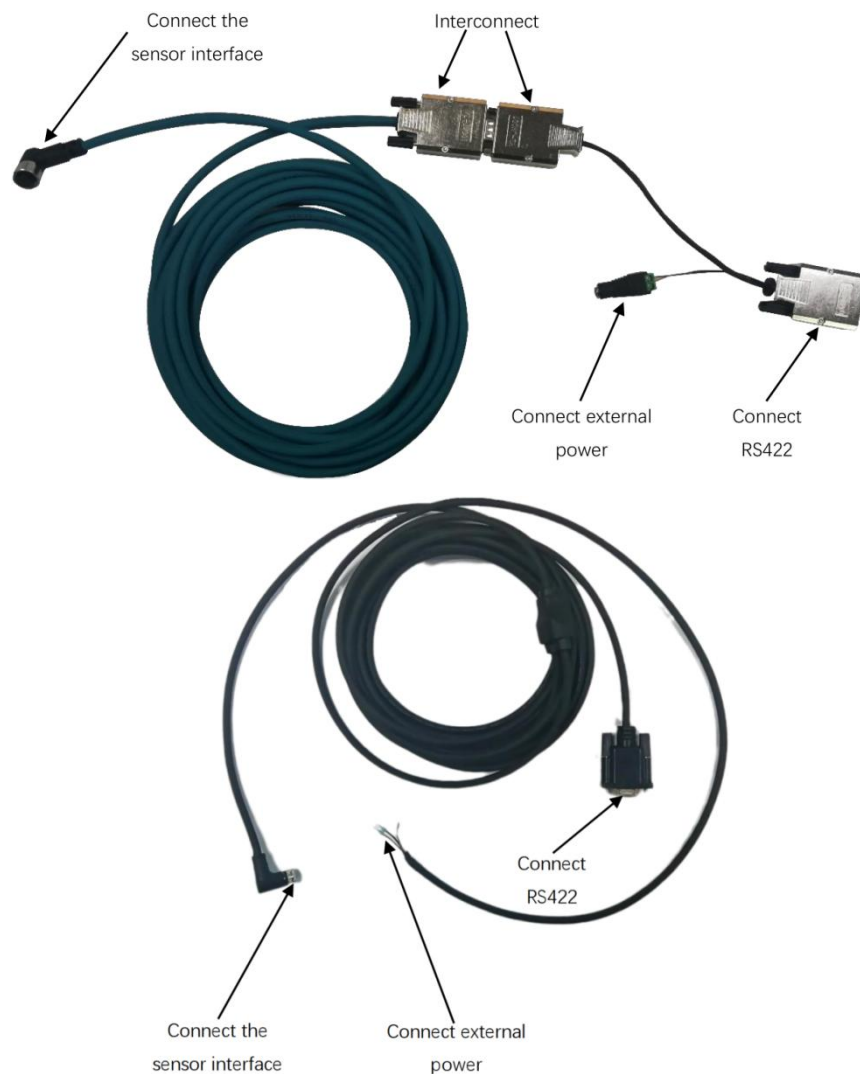


Fig. 3.2: Cable Installation for RS422

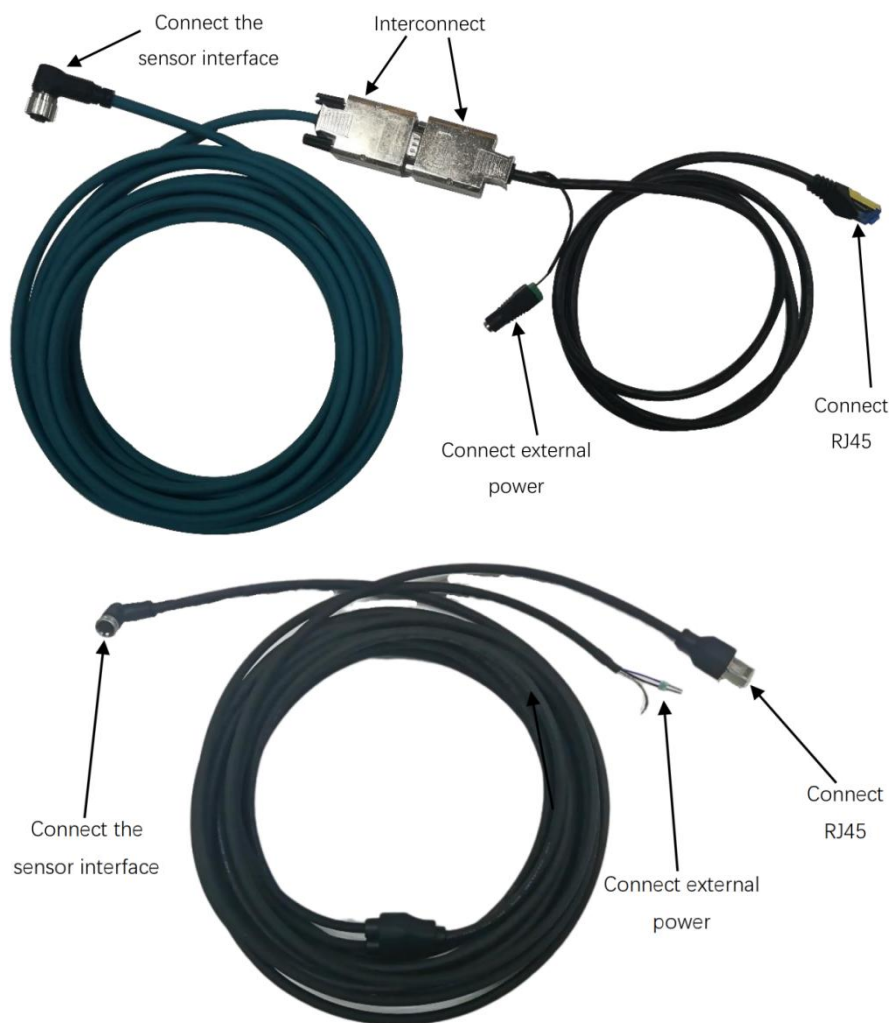
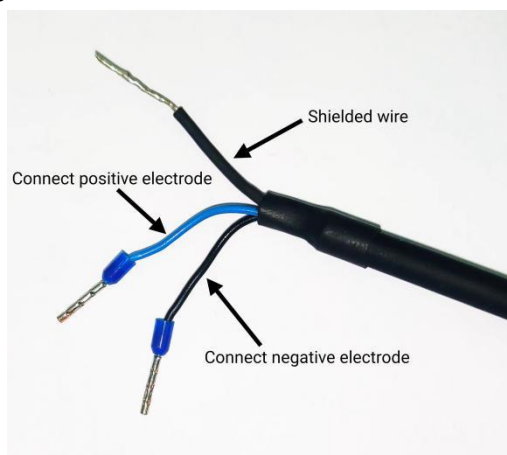


Fig. 3.3: Cable Installation for EtherCAT/Ethernet



Blue wire connected to positive electrode, black wire connected to negative electrode, shielded wire grounded

Fig. 3.4: Power cable definition

3.2 Cable Routing

The routing and radius of the bent cable are dependent on the user's circumstance. Unlike static condition (the cable is restricted in a static condition), the cables move forward and backward by force in dynamic condition. The cables are not only stretched out, twisted by force but also squeezed by fixed restriction, which make the cables deformed and damaged (including physic and inner insulation destruction) and jeopardize the equipment.

Reasonable restraint to movement of cables and careful planning of path will keep good communication and extend connector life.

NOTICE: The maximum length of cable is 30 m.

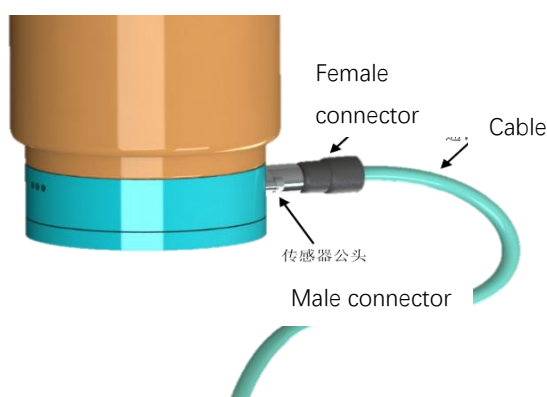


Fig. 3.5: Wiring Drawing



CAUTION: When routing cables do not bend less than the minimum bending radius specified in Table 3.1. A bend radius too small causes the cable to fail from fatigue of the robot's repetitive motion.



CAUTION: The cable must withstand the repetitive motions of the robot without failing. The electrical line, especially where attached to the sensor's connector, must be routed to avoid stress failure, sharp bends, or a disconnection from the equipment. Improper routing may cause poor functionality of critical electrical lines, injury to personnel, or damage to equipment. Damage to the sensor or cable from improper routing will void the warranty. If the application results in cable rubbing, use a loose, plastic spiral wrap for protection.

Table 3.1 Bending Radius and Dynamic Twist Angle of Cable

Model Number	Description	Cable Diameter (mm)	Static Bending Radius (at ambient temperature) (mm)	Dynamic Bending Radius (at ambient temperature) (mm)	Dynamic Twist Angle per Unit Length
S/FTP CAT.6	Power and Transport	6.5	8*D	10*D	180°/m

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Notes:

1. Dependent on temperature of environment, the bending radius of cable is flexible. It's recommended to increase minimum dynamic bending radius reasonably at lower temperature.
2. Never force the cables excessively, which may damage the cables. The cables near the sensor shall be free of stress, pull, twist, cutting and other damage. A soft jacket can be used to cover the cable where the out layer is damaged.

3.3 Connect sensor to robot

Installation: Refer to drawing

Tools required: 4 mm hex wrench

Additional requirement: Clean environment, Loctite® 242 glue.

Installation procedure:

1. Before installation, check the connection joints of sensor and robot for any foreign substance.
2. Screw using 4mm hex wrench to connect the sensor to the arm of robot. tighten screws in 6N • m.
 - a. Minimum thread engagement length is 4.5, maximum thread engagement length is 6.5 mm.
 - b. If the sensor is used in vibrational environment, the glue can be applied to screws.
3. After the installation of sensor on the robot is completed, connect user's tool fixture on the sensor.
 - a. The minimum thread engagement length is 4.5 mm, maximum thread engagement length is 5.5 mm.
 - b. If the sensor is used in vibrational environment, the glue that secures fasteners can be used to ensure tight connection of tool to the sensor.
4. Connect the sensor and user's port via cable S/FTP CAT.6.
 - a. Connect a power cable. LED will start self-test while powered on.
5. After installation is complete, it's necessary to calibrate the sensor. The sensor can be put in use once verification is completed.

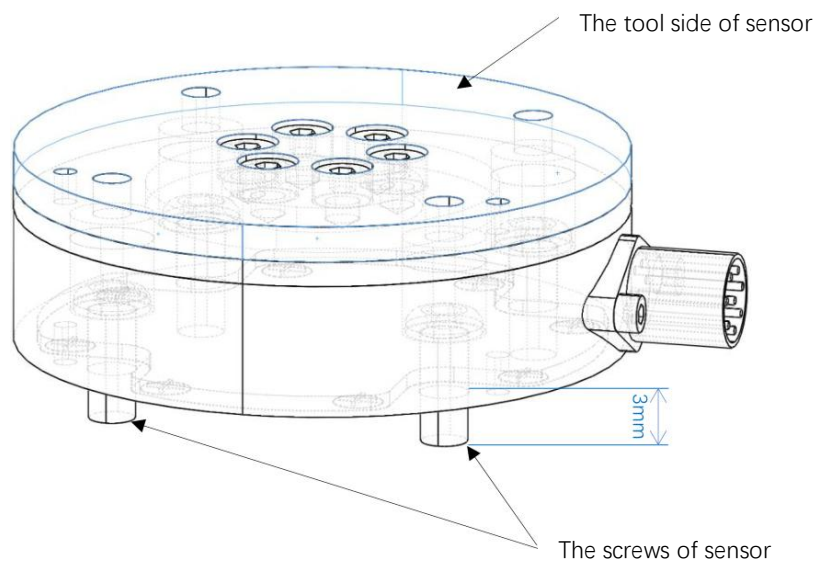


Fig. 3.6: The Perspective Drawing of Sensor



As shown in fig. 3.6, the ends of four screws (about 3mm in length) can be out of sensor. If the sensor is installed improperly, the heads of screws may attach the tool end and may seriously damage the sensor.

Suggestion: Don't screw individual screw completely. It's necessary to screw four screws in at least 3mm depth at first (Normally screw each screw at least five rounds. Stop screwing the screw while a slight resistance is met and screw another one) and then tighten screws at torque of 6N.m.

3.4 Dismount of the Sensor

Tools required: 4 mm hex wrench

1. Turn off all the power
2. Remove S/FTP CAT.6 cable from the sensor connector.
3. Remove the user's tool from the sensor.
4. Hold the sensor by hand, screw out screws using a hex wrench and dismount the sensor.

CAUTION: Ensure the cable is properly grounded. Improper shielding on the cables can cause communication errors and ineffectiveness of the sensors.



As shown in fig. 3.6, the ends of four screws (about 3mm in length) can be out of sensor. If the sensor is dismounted improperly, the heads of screws may attach the tool end and may seriously damage the sensor.

Suggestion: Don't unscrew individual screw completely. It's necessary to unscrew four screws in at least 3mm depth at first (Normally screw each screw at least five rounds. Stop screwing the screw while a slight resistance is met and screw another one) and then unscrew all the screws.

3.5 Sensor output signal distribution

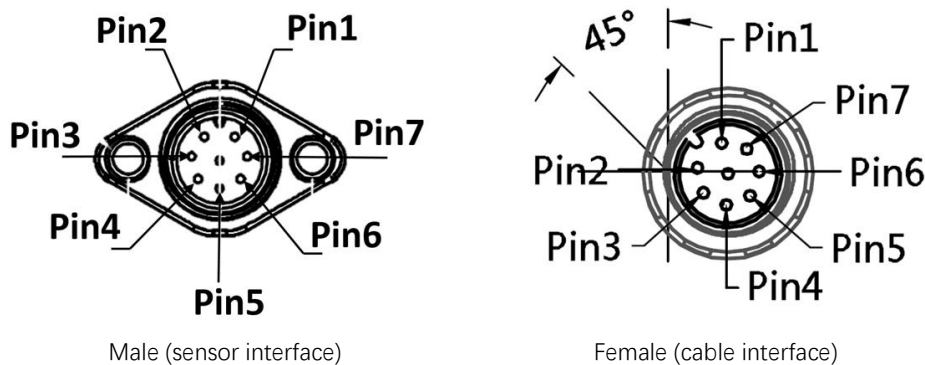


Fig. 3.7: The Connector Pins

The table 3.2 shows the numbers of pin corresponding to each internal signal.

Table 3.2 The output signals of sensor

Pins for S/FTP CAT.6			
Pin Number	Signal	Pin Number	Signal
1	V-	5	NA
2	Tx-	6	Rx-
3	Tx+	7	V+
4	Rx+	8	NA

3.6 Accuracy Verification Procedure

At the time when a sensor is initially used or a maintenance is carried out periodically, it's necessary to verify the accuracy of sensor prior to use.

1. Connect sensor to a weight which mass is known
2. Move the sensor to the following positions, Record the steady output of Fx, Fy and Fz at each point:

Point 1: +Z up

Point 2: +X up

Point 3: +Y up

Point 4: -X up

Point 5: -Y up

Point 6: -Z up

3. Calculate the mean value of following components:

$$F_{X.average} = (F_{x1} + F_{x2} + \dots + F_{x6}) / 6$$

$$F_{Y.average} = (F_{y1} + F_{y2} + \dots + F_{y6}) / 6$$

$$F_{Z.average} = (F_{z1} + F_{z2} + \dots + F_{z6}) / 6$$

4. Calculate the value of following components at each point.:

$$F_X = F_{Xn} - F_{X.average}$$

$$F_Y = F_{Yn} - F_{Y.average}$$

$$F_Z = F_{Zn} - F_{Z.average}$$

$$G = \sqrt{F_X^2 + F_Y^2 + F_Z^2}$$

5. S = Max G - Min G

$$S < 36n$$

The value of S shall be less than 36N each time when accuracy verification is carried out.

3.7 Nominal Moment of the Sensor



The sensor may be damaged if the actual moments exceed its nominal moments (in term of the M_x , M_y and M_z). The moment is a product of the force and the length of arm. If any one of values is excessively higher, the value of moment may be also excessively

Suggestions:

As shown in fig. 3.8, as a force of 5kg is loaded to the sensor at the distance of 0.2 meter away from the sensor, the maximum moment that the sensor would receive can be approximately calculated as $5\text{kg} \times 10\text{N/kg} \times 0.2\text{m} = 10\text{N.m}$. If the force is more distant from the sensor (i.e. the force is farther away from the sensor), or the tool receives additional force (e.g. grind workpiece using this tool), the maximum moment may exceed 10 N.m and the sensor which nominal moment is 10N.m may be damaged.

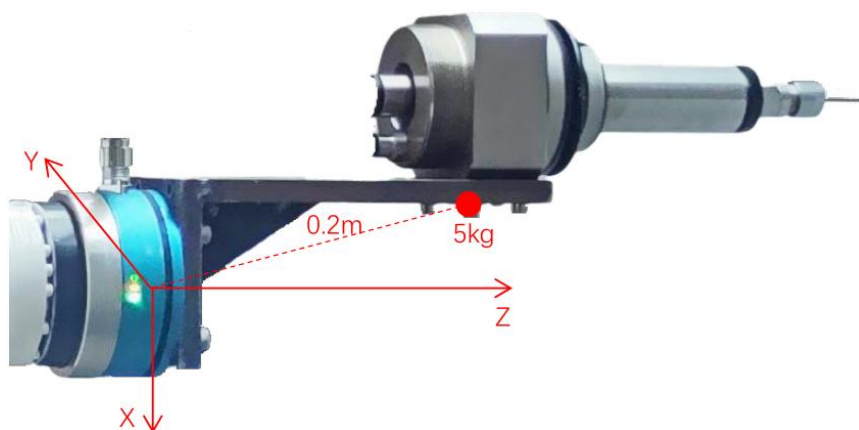


Fig. 3.8: Moment drawing

As shown in fig. 3.9, as a force of 10kg is loaded to the sensor at the distance of 0.1 meter away from the sensor, the maximum moment that the sensor would receive can be approximately calculated as $10\text{kg} \times 10\text{N/kg} \times 0.1\text{m} = 10\text{N.m}$. If the force is more distant from the sensor (i.e. the force is farther away from the sensor), or the tool receives additional force (e.g. pick up workpiece using this tool), the maximum moment may exceed 10 N.m and the sensor which nominal moment is 10N.m may be damaged.

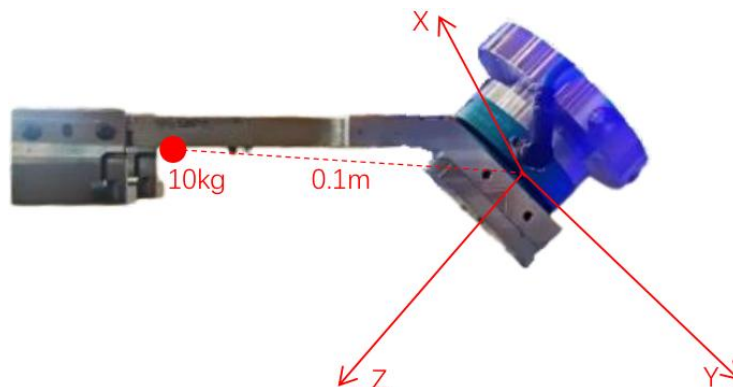


Fig. 3.9: Moment drawing

4. Operation Procedure

This section provides instruction to user the operation procedure that the user shall follow. To interact with the sensor, the operator shall be proficient with RS422 port, EtherCAT bus, Ethernet UDP or Modbus TCP protocol.

4.1 Sensor Environment

To ensure the life time and accuracy of the sensor, the sensor's environment must conform to its IP rating. This product has a protection rating of IP65.

CAUTION: *In case of damage of sensor jacket, the moisture or water may come into the sensor. Always check cable jacket to make sure they are in good condition.*

NOTICE: Strong and alternative electromagnetic fields may interference with the sensor, e.g. magnetic field generated by magnetic resonance imaging (MRI) machines.

The sensor is started as the power is switched on. There is no response in seconds until its status is stabilized since the sensor is started. We recommend to input commands to sensor 5 seconds later.

After the sensor is started, the output may fluctuate with internal temperature. To stabilize the output of sensor. We recommend to preheat the sensor for 10-30 minutes prior to use.

Device sampling rate: 1000HZ

Low-pass Filter: 500HZ

4.2 LED Function

Network indicator (L/A): it turns to 'GREEN' and blinks as ethernet communication cable is connected. It turns to 'OFF' as communication ethernet cable is disconnected.

Slave status indicator (Status): It's always 'OFF' for EtherCAT communication as slave is in status of Init/Bootstrap. it turns to 'YELLOW' and quickly blinks as slave is in the status of Pre-Op . It turns to 'YELLOW' and slowly blinks as slave is in the status of Safe-Op.. It turns to 'YELLOW' and 'ON' as slave is in the status of Op. It turns to 'OFF' for other types of communications.

Sensor working status indicator (P/E) : It turns to 'YELLOWGREEN' and 'ON' in normal condition.

4.3 RS422 Bus

The host computer interacts with the sensor using standard RS422 communication protocol. The baud rate is 921600bps, 8 bit data bit. 1 bit stop, no check bit.

As the sensor is powered on, it's defaulted as continuous sending status.

4.3.1 Communication content

Command	Sender	Receiver	Sending frequency
Data acquisition	Sensor	Host computer	1000Hz
Stop sending data	Host computer	Sensor	trigger
Start sending data	Host computer	Sensor	trigger

4.3.2 Communication protocol

1. Data acquisition command

Byte Number	Byte Content	Description
1	0xA5	Header
2~5	Time	float, Low byte first
6~9	1 Channel data	float, Low byte first
10~13	2 Channel data	float, Low byte first
14~17	3 Channel data	float, Low byte first
18~21	4 Channel data	float, Low byte first
22~25	5 Channel data	float, Low byte first
26~29	6 Channel data	float, Low byte first
30~33	Temperature	float, Low byte first
34~35	Reserved	
36	0x0D	suffix

2、Stop-sending-data command

Byte Number	Byte Content	Description
1	0xC0	Header
2	0xF8	
3	0xF8	
4	0xC0	suffix

3、Data sending command

Byte Number	Byte Content	Description
1	0xC0	Header
2	0xF7	
3	0xF7	
4	0xC0	suffix

4.4 EtherCAT communication

PDO interface

TxPDO

Subindex	Name	Type	Description
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0x01	Fx	DINT	Force component along X axis, 10000 corresponding 1N
0x02	Fy	DINT	Force component Y axis, 10000 corresponding 1N
0x03	Fz	DINT	Force component Z axis, 10000 corresponding 1N
0x04	Mx	DINT	Moment component, Mx, 10000 corresponding 1N.m
0x05	My	DINT	Moment component, My, 10000 corresponding 1N.m
0x06	Mz	DINT	Moment component, Mz, 10000 corresponding 1N.m
0x08	Sample Counter	UDINT	Run time, 1 corresponding 1ms, Based on this value, it can be determined if the incoming data frame is same as the last data frame or not.
0x09	Temper	DINT	Temperature, 10 corresponding 1℃

RxPDO

Subindex	Name	Type	Description
0x01	Control Code	DINT	Clearance of data at the uprising time from zero bit

4.5 Ethernet UDP communication

4.5.1 Sensor information

Data sending frequency: 100Hz / 500Hz / 1000Hz

IP address: 192.168.1.20

UDP port: 49152

Sensor: Server

The sensor will keep monitoring 49152 port for UDP input request, execute corresponding functions and respond after the sensor is powered on.

4.5.2 Communication protocol

1. Description of data requesting command

Command	Code	Function description	Output
Stop	0x0000	Stop data output	NA
Data request	0x0001 or 0x0002	Output a single data set while count is 1 or multiple data set while count > 1 or output continuously responsive data package while count is zero.	Responsive data package
Set zero point	0x0042	Set zero point	NA

2. Format of data requesting command

```
typedef struct _req_struct
```

```
{
```

```
    unsigned short magic;      // assigned as 0x1234
    unsigned short command;    // command to be executed as fore-mentioned
    unsigned int count;        // count, 0 represents continuous output
} reqStruct;
```

magic takes 2 bytes, assigned 0x1234, for identification of request command

command takes 2 bytes, its definition is shown in Table 1.1

count takes 4 bytes, for assignment of number of responsive data package to output, especially, zero represents unlimited number. Sensor will output responsive data package continuously until stop command is received.

3. Format of data responding command

typedef struct _resp_struct

```
{
    unsigned int rdt_sequence;      // count of response
    unsigned int ft_sequence;      // Count of data request
    unsigned int status;           // Sensor status
    int Fx;                        // Force component in x axis, unit: N/1000000
    int Fy;                        // Force component in Y axis, unit: N/1000000
    int Fz;                        // Force component in Z axis, unit: N/1000000
    int Tx;                        // Moment component around X axis, unit: N*m/1000000
    int Ty;                        // Moment component around Y axis, unit: N*m/1000000
    int Tz;                        // Moment component around Z axis, unit: N*m/1000000
} respStruct;
```

rdt_sequence takes 4 bytes, low byte first, representing serial number of data response. It will plus 1 automatically every time while sensor sends data response. *rdt_sequence* will be reset to '1' while sensor receives a new data response, which means *rdt_sequence* is an increasing serial number of data response each time while data is requested. The host computer is able to determine whether there are any data is missing based on its value .

ft_sequence takes 4 bytes, low byte first, representing serial number of data response. *ft_sequence* will plus one automatically each time while the sensor gets a data set. When the value of *ft_sequence* is 0xFFFFFFFF, it will become zero after its value plus 1, thereafter it will increase continuously.

Each variable Fx,Fy and Fz takes 4 bytes, low byte first, representing the value of forces in N/1000000 unit in x, y and z axis respectively.

Each variable Tx,Ty and Tz takes 4 bytes, low byte first, representing the value of moments in N*m/1000000 unit around x, y and z axis respectively.

4.6 Modbus TCP communication

4.6.1 Configuration information

IP address: 192.168.0.20

Port number: 502

Sensor: Server(Slave)

Data transmission frequency: 1000Hz

4.6.2 Function description

After the sensor is powered on, it will continue to listen to Modbus TCP input requests on port 502, perform corresponding functions, and respond.

4.6.3 Communication protocol

1. Instruction description

Instruction	Code	Register address	Function description	Response
Read holding register	0x03	0x0000	Read force and moment data	Force and moment data package
		0x1088	Read sensor system information	Sensor information packet
Write single register	0x06	0x1000	Set zero point: write 1 to register on 0x1000	Request package
		0x1078-0x107F	Write sensor system information in stages. From 0x1078 to 0x107F, a total of 8 registers, 16 bytes. Each register is 2 bytes and 8 registers are written separately.	Request package
Write multiple registers	0x10	0x1078	Write sensor information	Response package

Description:

1. There are 8 registers from 0x1078 to 0x107F, 16 bytes. Used to store the input sensor system information.
2. There are 8 registers from 0x1088 to 0x108F, 16 bytes. Used to store the output sensor system information.

3. Sensor system information needs to be written first and then read out. That is, write 0x1078 to 0x107F in sequence, then read the 8 registers starting at 0x1088.

4. The sensor system information can be written by calling the single register instruction (0x06) multiple times or by writing multiple register instructions (0x10) at one time. It is recommended to use the write multiple register instruction.

5. The force and moment data obtained by the above instructions are all 16-bit unsigned integers, low byte first, which need to be converted to decimal data first, and then the actual force and moment value can be obtained by the following calculation:

Actual force value (N) = (decimal data - 32768) / 32.768

Actual moment value (N·m) = (decimal data - 32768) / 327.68

2. Command example

(1) Write sensor system information (0x06)

Send: 00 00 00 00 00 06 00 06 10 78 0F B0

Respond: 00 00 00 00 00 06 00 06 10 78 0F B0

Send: 00 00 00 00 00 06 00 06 10 79 3F 10

Respond: 00 00 00 00 00 06 00 06 10 79 3F 10

Send: 00 00 00 00 00 06 00 06 10 7A D9 40

Respond: 00 00 00 00 00 06 00 06 10 7A D9 40

Send: 00 00 00 00 00 06 00 06 10 7B A3 01

Respond: 00 00 00 00 00 06 00 06 10 7B A3 01

Send: 00 00 00 00 00 06 00 06 10 7C 7C 68

Respond: 00 00 00 00 00 06 00 06 10 7C 7C 68

Send: 00 00 00 00 00 06 00 06 10 7D 34 BA

Respond: 00 00 00 00 00 06 00 06 10 7D 34 BA

Send: 00 00 00 00 00 06 00 06 10 7E 49 CE

Respond: 00 00 00 00 00 06 00 06 10 7E 49 CE

Send: 00 00 00 00 00 06 00 06 10 7F E8 2F

Respond: 00 00 00 00 00 06 00 06 10 7F E8 2F

(2) Write sensor system information (0x10)

Send: 00 00 00 00 00 16 00 10 10 78 00 08 0F B0 3F 10 D9 40 A3 01 7C 68 34 BA 49 CE E8 2F

Respond: 00 00 00 00 00 06 00 10 10 78 00 08

(3) Read sensor system information (0x03)

Send: 00 00 00 00 00 06 00 03 10 88 00 08

Respond: 00 00 00 00 00 13 00 03 10 93 3F 7B AC 8C 19 6F 6D 06 5F CA A6 70 3A 16 3D

(4) Set zero point (0x06)

Send: 00 00 00 00 00 06 00 06 10 00 00 01

Respond: 00 00 00 00 00 06 00 06 10 00 00 01

(5) Read sensor force and moment data (0x03)

Send: 00 00 00 00 00 06 00 03 00 00 00 0C

Respond: 00 00 00 00 00 1B 00 03 18 E6 3C 00 00 FB BB 00 00 F5 FE 00 00 00 1D 00 00 02 2E
00 00 00 36 00 00

5. Maintenance

5.1 Periodic Inspection

With industrial-type applications that frequently move the system's cabling, you should check the cable jacket for signs of wear. The WRIST is IP65 rated and must be kept free of moisture. Debris and dust should be kept from accumulating on or in the sensor. The surface of the sensor can be cleaned with isopropyl alcohol, if contaminated by its environment. The sensor itself should experience no wear, if used within the operating ranges and fastened to the proper moment specifications.

5.2 Periodic Calibrating

For schools, research institutes, etc., where the accuracy of the product is extremely high., Users need to follow applicable ISO-9000-type standards for calibration. For the field where the product use environment is harsh, to ensure the accuracy of the product, we recommend annual accuracy checks

5.3 Troubleshooting

The information in this section should answer many questions that might arise in the field. Customer service is available for problems or questions not addressed in the manuals.

The most common fault of a sensor is reading error in Force and Moment., Inaccurate data from the transducer's strain gages can cause errors in force/moment readings. These errors can result in problems with transducer biasing and accuracy.

Listed in the following table are the basic problems of inaccurate data:

Question/Problem	Answer/Solution
Noise	Jumps in force moment data readings (with the sensor unloaded) greater than 0.1% of full scale counts is abnormal. Noise can be caused by mechanical vibrations and electrical disturbances, possibly from a poor ground. Noise can also indicate component failure within the system. Make sure that the DC supply voltage for the WRIST sensor has little to no noise superimposed. The sensor should be grounded through installation construction

Drift	After a load is removed or applied, the raw gage reading does not stabilize but continues to increase or decrease. A shift in the raw gage reading is observed more easily in the resolved data mode using the bias command. Some drift from a change in temperature or mechanical coupling is normal. Mechanical coupling occurs when a tool plate contacts the sensor body, for example, debris between the tool adapter plate and the sensor body or in applications such as hoses and wires attached to a tool.
Hysteresis	When the sensor is loaded and then unloaded, gage readings do not return quickly and completely to their original readings. Hysteresis is caused by mechanical coupling or internal failure. A certain range of hysteresis is normal.
Sensor not streaming measurement data to the customer devices	Verify the sensor is correctly installed. Ensure the robot mounting and tool adapter plates are installed on the proper side of the sensor

6. Product Specifications

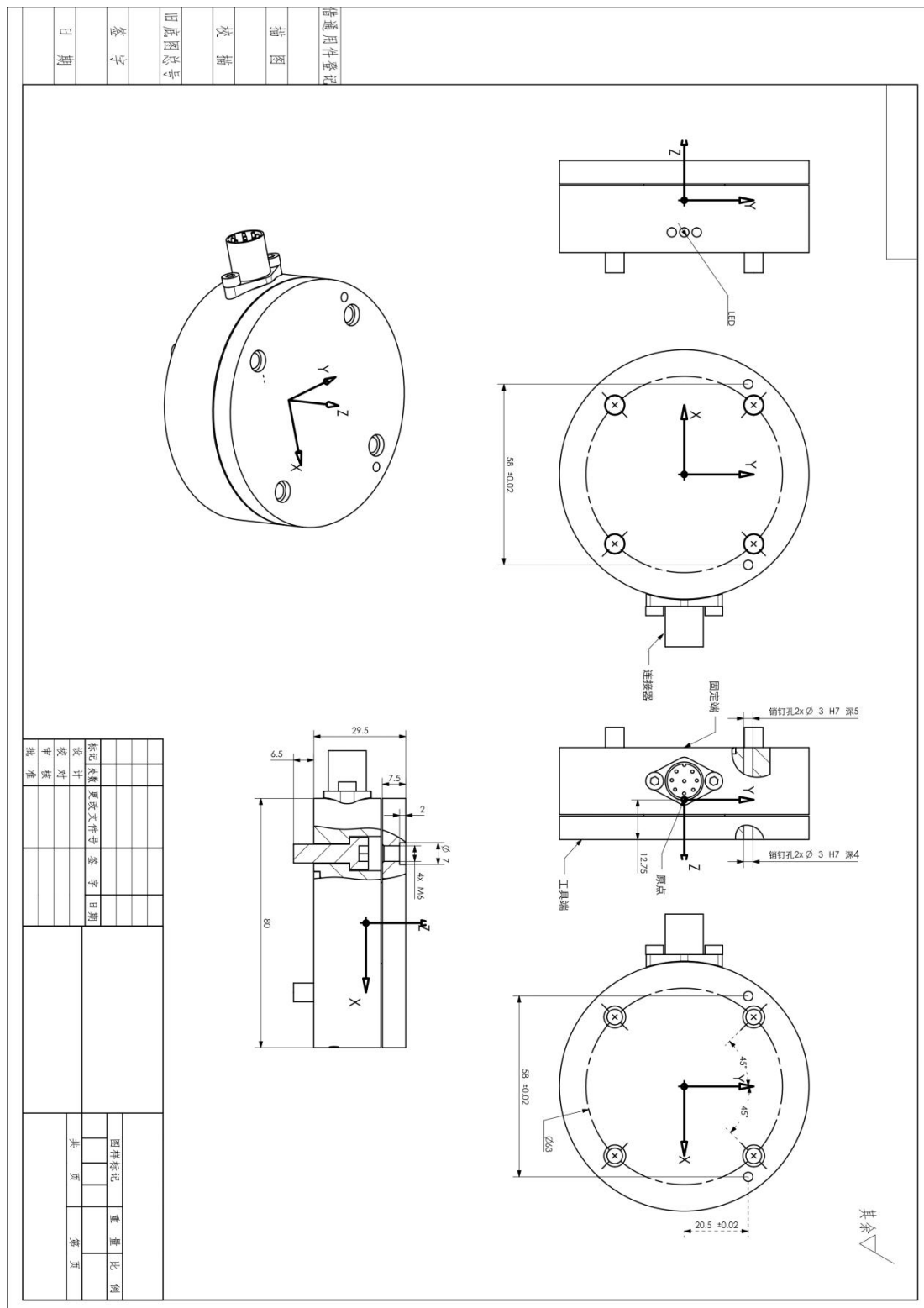
6.1 Storage and Operating Conditions

Environmental conditions	
Parameter	Value
Storage temperature, °C	-20~85
Operating temperature, °C	-10~60
Relative humidity	<95%, non-condensing
Storage vibration	5g
Operating vibration	2g
Corrosive liquids or gases	No protection

6.2 Electrical Specifications

Sensor		Model			
		ST-6-170-6	ST-6-200-10	ST-6-500-20	ST-6-1500-40
Minimal Load	Fx, Fy (N)	170	200	500	1500
	Fz (N)	350	400	900	3000
	Mx, My, Mz (Nm)	6	10	20	40
Resolution	Fx, Fy (N)	0.1	0.1	0.2	0.5
	Fz (N)	0.2	0.2	0.3	0.8
	Mx, My, Mz (Nm)	0.005	0.005	0.007	0.021
Overload		5 times	5 times	5 times	5 times
Accuracy (include delay, linearity, coupling, and drift)		1.2%FS	1.2%FS	1.2%FS	1.2%FS
Size	mm	φ80×H25.5	φ80×H25.5	φ80×H25.5	φ 80×H25.5
Sampling Rate	Hz	1000	1000	1000	1000
Communication		RS422/485 EtherCAT Ethernet	RS422/485 EtherCAT Ethernet	RS422/485 EtherCAT Ethernet	RS422/485 EtherCAT Ethernet
Color		Light blue	Light blue	Light blue	Light blue
Quality	g	Ca. 260	Ca. 260	Ca. 280	Ca. 620
Protection Level		IP65/IP67	IP65/IP67	IP65/IP67	IP65/IP67
Input Voltage	V	DC 9-36	DC 9-36	DC 9-36	DC 9-36
Temperature Compensation		Built-in	Built-in	Built-in	Built-in
Power	W	≤2	≤2	≤2	≤2

7. Drawings



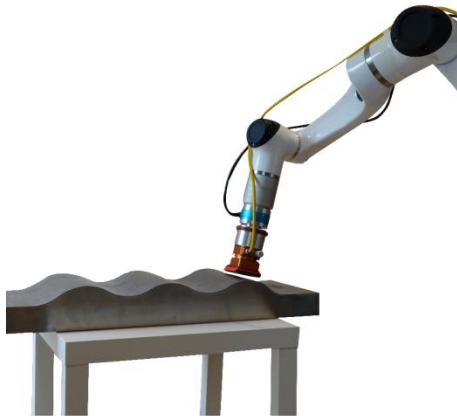
8. Application Examples



Example 1: Edging and Chamfer



Example 2: Drill



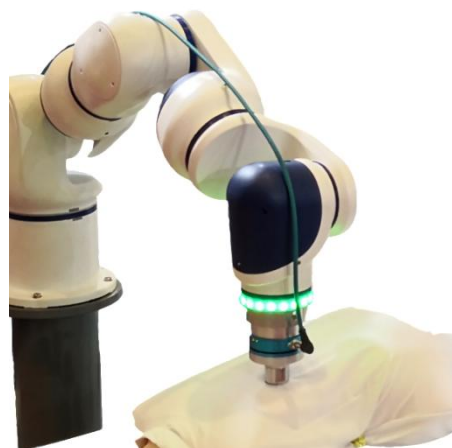
Example 3: Polish



Example 4: Flexible Assembly



Example 5: Collision Checking



Example 6: Medical Massage

更改记录

时间	版本	更改内容	更改人
2020.07.06	1.5.2	1、 更新 4.6 Modbus TCP 通讯	付伟宁
2020.08.19	1.5.3	1、 更新首页产品图片 2、 删除 3.5 中关于线色的描述	付伟宁
2020.08.26	1.5.4	1、 新增 4.5.1 中传感器为 server 2、 更新 4.5.2 中关于单位的描述 3、 更新 6.2 产品型号参数	付伟宁
2020.12.02	1.5.5	1、 更新电话号，新增网址 2、 更新 4.2 中 L/A 状态灯的描述 3、 更新 4.5 和 4.6 中大小端的描述 4、 更新第七章图纸的销钉孔深度	付伟宁